SHREYA THAKUR

+1-469-916-8196| thshreya7code@gmail.com | github.com/shreya7code | linkedin.com/in/shreya-thakur7/ | Portfolio | NY, USA (Open to All)

Software Engineer with 4+ years of experience in Python backend and AI/ML systems, specializing in LLM applications, semantic search, and cloud-native architectures on AWS and Azure. Experienced in building scalable, production ML solutions with strong MLOps practices.

WORK EXPERIENCE

Stealth | Full Stack Developer | CA, USA *Aug 2025 – Present*

- Engineered **Python** and **Django** backend services for client-specific **LLM-**based AI Assistants, using **FastAPI** async endpoints for model inference and **PostgreSQL** for application data, improving scalability and response efficiency by 40%.
- Developed zero-shot intent routing with **Hugging Face** BART-MNLI Transformers for **JWT/OAuth2** multi-tenant clients, raising accuracy 25%.
- Integrated **RAG** using **Pinecone, LangChain**, & **LangGraph**, with **Ollama** on-device inference for privacy, reducing external API costs by **30%**.
- Built **OWASP-**aligned LLM guardrails for prompt injection, PII redaction, with **LLM-as-judge** and **red-teaming**, detecting ~95% of attacks.
- Tracked p95/p99 latency & TTFT in **Prometheus** & **Grafana** with LLM evals (Recall@K, groundedness), fixing a production latency regression.
- Containerized backend & ML services using **Docker**, **Kubernetes** & **GitHub Actions CI/CD**, reducing setup by 70% with AWS EC2 deployments.

University at Buffalo | Graduate Teaching Assistant | NY, USA *Sep 2024 – Jun 2025*

- Mentored 200+ students in **deep learning** and **Hadoop** labs, covering model evaluation, buffer overflow**,** SQL injection**,** and cryptography**.**
- Supported applied research on an end-to-end document intelligence system for OCR, summarization, translation, and TTS, integrating **mT5, NLLB**, and **FastText** language models.
- Architected a custom OCR pipeline using **PyTorch**, **ResNet**, **CRNN**, **BiLSTM**, and **CTC**, tracking experiments in **MLflow** and selecting checkpoints by CER to achieve 92% accuracy and a 35% CER reduction; fine-tuned **TrOCR** for handwritten text.
- Containerized the service with **Docker**, enabled GPU inference, and deployed it on Azure using **Azure Container Registry** and **App Service**.

Wipro Ltd. | Software Engineer | Kolkata, India *Aug 2019 – Aug 2022*

Project: British Petroleum (BP)

- Programmed an AI-powered semantic search platform using **Python, Azure AI Search** (Cognitive Search), **NLP**, and a **React** frontend, indexing 10M+ documents, improving search precision by 25% and reducing query latency by 40%.
- Developed an ML-based PII detection using Flask, Presidio, and event-driven Azure Functions, reducing compliance effort by 80%.
- Built an NLP preprocessing microservice with **Python** and **NLTK** for Azure AI entity recognition, improving ML accuracy by 20% via tokenization and normalization.
- Engineered **Azure Data Factory** and **PySpark** pipelines on **Databricks** and **SQL**, delivering an audit analytics dashboard in 10 days, processing 10TB+ with 35% lower latency.
- Improved CI/CD with **Docker, Azure DevOps**, and **Key Vault** in **Agile** sprints, cutting deployment failures by 30% and improving reliability.

Project: Cisco Partner Portal

- Implemented scalable backend services using **Java, SQL, and REST APIs**, supporting partner onboarding and reducing API latency by **35%**.
- Refactored backend modules using **object-oriented design**, improving maintainability and delivery in distributed environments by **40%**.
- Automated functional and integration testing using **Selenium and Cucumber**, achieving **95% test coverage** and reducing regressions.
- Integrated CI/CD pipelines with **Jenkins and Docker**, reducing release cycles by **60%** and enabling faster delivery of backend features.

PROJECTS

AgenticAI Model Safety Evaluation Platform [Link] *Jan 2026*

- Worked on agentic AI safety & evaluation Platform to automate LLM **red-teaming** and regression testing across 1,500+ test cases, reducing manual review by 90% and enabling quality monitoring through **Grafana** dashboards.
- Orchestrated a 4-agent **LangGraph** pipeline with **TypeScript**, **pgvector**, and **PostgreSQL** for **A/B model** comparison, automated scoring, and regression detection for quality drops above 5%.

Smart Infra Monitor – Real-Time Infra Monitoring & Anomaly Detection [Link] *Jul 2025*

- Improved ML anomaly detection, processing **2K+ infrastructure events/second** through scalable **Kafka, FastAPI,** and **InfluxDB** pipelines.
- Deployed containerized services on **AWS (EC2, Docker)** with rule-based and ML detection, improving monitoring efficiency by **30%**.

SKILLS

Programming & Backend: Python, Java, C++, TypeScript, SQL, FastAPI, Flask, REST APIs, GraphQL
Machine Learning & Deep Learning: Scikit-learn, Pandas, NumPy, Feature Engineering, TensorFlow, PyTorch, CNN, CRNN, BiLSTM, CTC, NLP, Hugging Face, Transformers, LoRA, SHAP, LIME, MLflow
Agentic AI & LLMs: LangChain, LangGraph, LLMs, Multi-Agent Orchestration, RAG, Prompt Engineering, Evaluation, LLM-as-Judge, Pinecone
Data Engineering & Analytics: SQL, PostgreSQL, Data Modeling, ETL Pipelines, PySpark, Apache Spark, Airflow, Snowflake, Power BI, Tableau
Cloud & MLOps: AWS (EC2, S3, Lambda, SageMaker), Azure (Functions, Cognitive Search, Blob Storage), Docker, Kubernetes, CI/CD, Terraform
Distributed Systems & DevOps: Kafka, InfluxDB, Redis, Microservices, Event Driven Architecture, Monitoring & Logging, Jenkins, GitHub
Core Concepts: Data Structures & Algorithms, System Design, Model Optimization, Experiment Tracking, Model Deployment
Certifications: AWS Solutions Architect – Associate, Introduction to Generative (AWS), AI-900 (Microsoft), DBMS (NPTEL), Python (MTA)

EDUCATION

University at Buffalo, The State University of New York | MS (Masters): Computer Science and Engineering *Buffalo, NY*
West Bengal University of Technology | B.Tech (Bachelors): Information Technology *Kolkata, WB, India*